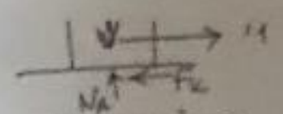
$F = 50\ N$

c-)



$+\uparrow \sum F_y = m.a_y;\quad N_A - 15.(9,81) = 0 \quad N_A = 147,15\ N$

$\overset{+}{\rightarrow} \sum F_x = m.a_x;\quad T_1 - 0,3.(147,15) = 15.a$



$+\uparrow \sum F_y = m.a_y \quad T_2 - 20.(9,81) = -20.a$

$a = r.\alpha \Rightarrow \alpha = \frac{a}{r} = \frac{a}{0,1} = 10.a$

$I_o = \frac{1}{2} m.r^2 = \frac{1}{2}.5.(0,1)^2 = 0,025\ kg.m^2$

$\sum M_o = I_o.\alpha \quad T_1.0,1 - T_2.(0,1) = -0,025.a$

$T_1 - 0,3.(147,15) = 15.a$

$T_2 - 20.(9,81) = -20.a$

$T_1 - T_2 = -2,5.a$

ortak çözülürse $a = 4,054\ m/s^2$

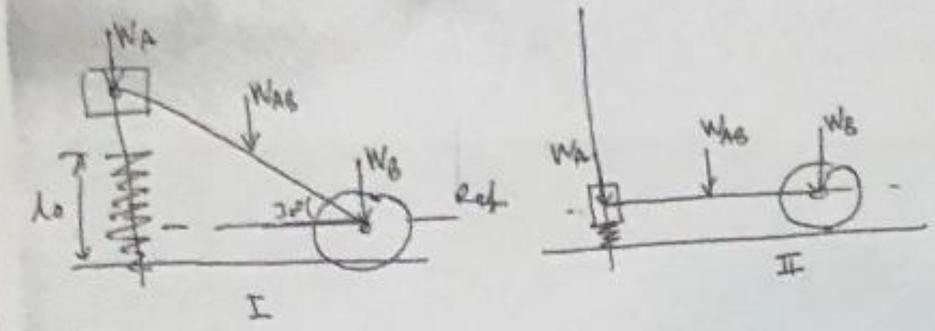
$T_1 = 104,967\ N$

$T_2 = 115,104\ N$

a-)

$V_I = W_A.y_{G_A} + W_{AB}.y_{G_{AB}} + 0$

$V_I = 19,62.0,375 + 49,05.0,188 = 16,58\ J$



$V_{II} = \frac{1}{2} k.\Delta x^2 = 0,5.100.(0,08)^2 = 0,32\ J$

$T_2 = 1,03.\omega_{AB}^2$

Enerjinin Korunumu

$\Delta T = T_{II} - T_I = 1,03.\omega_{AB}^2$

$\Delta V = V_{II} - V_I = 0,32 - 16,58 = -16,26\ J$

$0 = \Delta T + \Delta V$

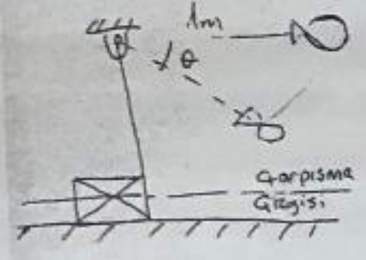
$0 = 1,03.\omega_{AB}^2 - 16,26$

$(\omega_{AB})_2 = 3,97\ rad/s$

$v^2 = v_0^2 + 2.a.\Delta x$

$v^2 = 0 + 2.4,054.2 \Rightarrow v = 4,03\ m/s \downarrow$

c.3)



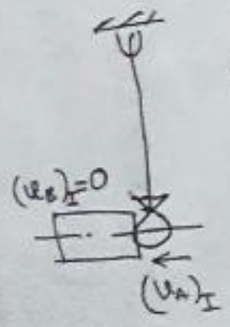
Bu problem merkezsel çarpışma problemi dir. Torbanın kutuya çarpmasından hemen önceki hızını hesaplamak gerekir.

Enerjinin Korunumu

$\theta = 0^\circ \quad T_o + V_o = T_I + V_I$

$0 + 0 = \frac{1}{2}\left(\frac{30}{9,81}\right).(v_A)_I^2 - 30.1$

$(V_A)_I = 4,43\ m/s$



Momentumun Korunumu

A ve B'nin çarpışmadan hemen sonra sola doğru gittiğini düşünürsek;

Sisteme momentumun korunumu

$m_B.(v_B)_I + m_A.(v_A)_I = m_B.(v_B)_2 + m_A.(v_A)_2$

$0 + \frac{30}{9,81}.4,43 = \frac{90}{9,81}.(v_B)_2 + \frac{30}{9,81}.(v_A)_2$

$(v_A)_2 = 4,43 - 3.(v_B)_2$

Geri Dönme Katsayısı

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1} \Rightarrow 0,5 = \frac{(v_B)_2 - (v_A)_2}{4,43 - 0}$

$(v_B)_2 = (v_A)_2 - 2,22$

$(v_A)_2 = 4,43 - 3.[(v_A)_2 - 2,22] \Rightarrow (v_A)_2 = -0,55\ m/s$

$= 0,55\ m/s \rightarrow$

$(v_B)_2 = 1,66\ m/s$

Enerji: Kaybı; çarpışmadan hemen sonra ve hemen sonrası iş ve enerji ilkesini uygularsak

$\Sigma U_{1\rightarrow 2} = T_2 - T_1 = \left[\frac{1}{2}\left(\frac{90}{9,81}\right).1,66^2 + \frac{1}{2}.\left(\frac{30}{9,81}\right).0,55^2\right] - \left[\frac{1}{2}\left(\frac{30}{9,81}\right).(4,43)^2\right]$

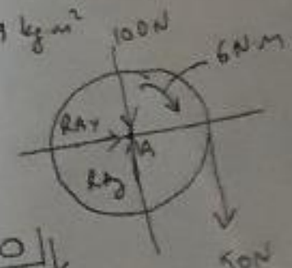
$= -16,9\ J$

c.4)

Disk in kütle merkezi hareket etmez; sadece yükleme diskin saat yönünde dönmesine neden olur

Diskin sabit dönme eksenine göre atalet momenti

$I_A = \frac{1}{2} m.r^2 = \frac{1}{2}.\left(\frac{200}{9,81}\right)(0,25)^2 = 0,319\ kg.m^2$



İmpuls ve Momentum ilkesi

$\overset{+}{\rightarrow} \quad m.(v_{A_x})_1 + \sum \int_{t_1}^{t_2} F_x.dt = m.(v_{A_x})_2$

$0 + R_{AX}.2 = 0 \Rightarrow R_{AX} = 0$

$+\uparrow \quad m.(v_{A_y})_1 + \sum \int_{t_1}^{t_2} F_y.dt = m.(v_{A_y})_2$

$0 + R_{A_y}.2 - 100.2 - 50.2 = 0$

$R_{A_y} = 150\ N$

$\overset{+}{\curvearrowright} \quad I_A.\omega_1 + \sum \int_{t_1}^{t_2} M_A.dt = I_A.\omega_2$

$0 + 6.2 + (50.0,25).2 = 0,319.\omega_2$

$\omega_2 = 116\ rad/s$

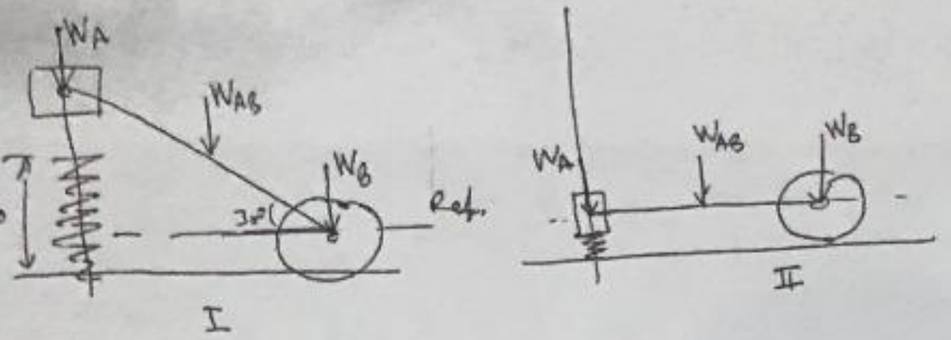
(İmpuls ve Momentum İlkesi) (25 puan)



a)

$V_I = W_A \cdot y_{GA} + W_{AB} \cdot y_{GAB} + 0$

$V_I = 19,62 \cdot 0,375 + 49,05 \cdot 0,188 = 16,58$ J



$V_{II} = \frac{1}{2} k \cdot \Delta x^2 = 0,5 \cdot 100 \cdot (0,08)^2 = 0,32$ J

$T_2 = 1,03 \cdot \omega_{AB}^2$

Enerjinin Korunumu

$\Delta T = T_{II} - T_I = 1,03 \cdot \omega_{AB}^2$

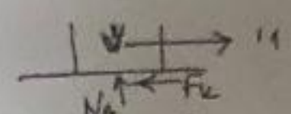
$\Delta V = V_{II} - V_I = 0,32 - 16,58 = -16,26$ J

$0 = \Delta T + \Delta V$

$0 = 1,03 \cdot \omega_{AB}^2 - 16,26$

$(\omega_{AB})_2 = 3,97$ rad/s

b)



$+\uparrow \sum F_y = m \cdot a_y$; $N_A - 15 \cdot (9,81) = 0$ $N_A = 147,15$ N

$\stackrel{+}{\rightarrow} \sum F_x = m \cdot a_x$; $T_1 - 0,3 \cdot (147,15) = 15 \cdot a$



$+\uparrow \sum F_y = m \cdot a_y$ $T_2 - 20 \cdot (9,81) = -20 \cdot a$

$a = r \cdot \alpha \Rightarrow \alpha = \frac{a}{r} = \frac{a}{0,1} = 10 \cdot a$

$I_o = \frac{1}{2} m \cdot r^2 = \frac{1}{2} \cdot 5 \cdot (0,1)^2 = 0,025$ kg.m²

$\curvearrowleft \sum M_o = I_o \cdot \alpha$ $T_1 \cdot 0,1 - T_2 \cdot (0,1) = -0,025 \cdot a$

$T_1 - 0,3 \cdot (147,15) = 15 \cdot a$

$T_2 - 20 \cdot (9,81) = -20 \cdot a$

$T_1 - T_2 = -2,5 \cdot a$

ortak çözülürse $a = 4,054$ m/s²

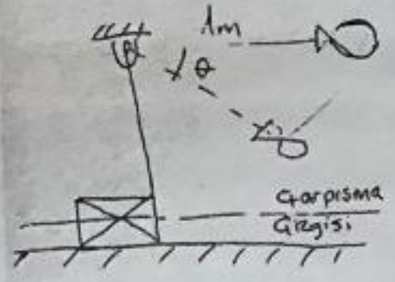
$T_1 = 104,967$ N

$T_2 = 115,104$ N

$v^2 = v_0^2 + 2 \cdot a \cdot \Delta x$

$v^2 = 0 + 2 \cdot 4,054 \cdot 2 \Rightarrow v = 4,03$ m/s ↓

c.3)



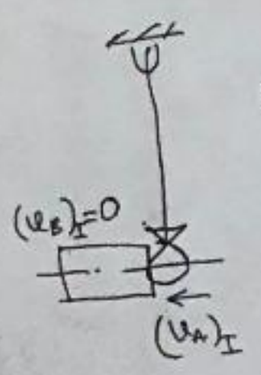
Bu problem merkezsel çarpışma problemi dir. Torbanın kutuya çarpmasından hemen önceki hızını hesaplamak gerekir.

Enerjinin Korunumu

$\theta = 0^\circ$ $T_o + V_o = T_I + V_I$

$0 + 0 = \frac{1}{2}\left(\frac{30}{9,81}\right) \cdot (v_A)_I^2 - 30 \cdot 1$

$(V_A)_I = 4,43$ m/s



Momentumun Korunumu

A ve B'nin çarpışmadan hemen sonra sola doğru gittiğini düşünürsek;

Sisteme momentumun korunumu

$m_B \cdot (v_B)_I + m_A (v_A)_I = m_B \cdot (v_B)_2 + m_A \cdot (v_A)_2$

$0 + \frac{30}{9,81} \cdot 4,43 = \frac{90}{9,81} \cdot (v_B)_2 + \frac{30}{9,81} \cdot (v_A)_2$

$(v_A)_2 = 4,43 - 3 (v_B)_2$

Geri Dönme Katsayısı

$e = \frac{(V_B)_2 - (V_A)_2}{(V_A)_1 - (V_B)_1} \Rightarrow 0,5 = \frac{(V_B)_2 - (V_A)_2}{4,43 - 0}$

$(V_B)_2 = (V_A)_2 - 2,22$

$(V_A)_2 = 4,43 - 3 \cdot [(V_A)_2 - 2,22] \Rightarrow (V_A)_2 = -0,55$ m/s

$= 0,55$ m/s ←

$(V_B)_2 = 1,66$ m/s

Enerji kaybı, çarpışmadan hemen sonra ve hemen sonrası ile ve enerji ilkesini uygularsak

$\Sigma U_{1-2} = T_2 - T_1 = \left[\frac{1}{2}\left(\frac{90}{9,81}\right) \cdot 1,66^2 + \frac{1}{2} \cdot \left(\frac{30}{9,81}\right) \cdot 0,55^2\right]$

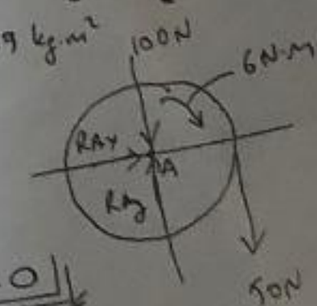
$- \left[\frac{1}{2}\left(\frac{30}{9,81}\right) \cdot (4,43)^2\right]$

c.4)

Disk kütle merkezi hareket etmez; ancak yüklenme diskin saat yönünde dönmesine neden olur

Diskin sabit dönme eksenine göre atalet momenti

$I_A = \frac{1}{2} m \cdot r^2 = \frac{1}{2} \cdot \left(\frac{200}{9,81}\right) (0,25)^2 = 0,319$ kg.m²



İmpuls ve Momentum ilkesi

$\stackrel{+}{\rightarrow}$ $m \cdot (v_{Ax})_1 + \sum \int_{t_1}^{t_2} F_x \cdot dt = m \cdot (v_{Ax})_2$

$0 + R_{Ax} \cdot 2 = 0 \Rightarrow R_{Ax} = 0$

$+\uparrow$ $m \cdot (v_{Ay})_1 + \sum \int_{t_1}^{t_2} F_y \cdot dt = m \cdot (v_{Ay})_2$

$0 + R_{Ay} \cdot 2 - 100 \cdot 2 - 50 \cdot 2 = 0$

$R_{Ay} = 150$ N

$\curvearrowright$ $I_A \cdot \omega_1 + \sum \int_{t_1}^{t_2} M_A \cdot dt = I_A \cdot \omega_2$

$0 + 6 \cdot 2 + (50 \cdot 0,25) \cdot 2 = 0,319 \cdot \omega_2$

$\omega_2 = 116$ rad/s

$\Delta V = V_2 - V_1 = 0,32 - 16,58 = -16,26$ J

$0 = \Delta T + \Delta V$

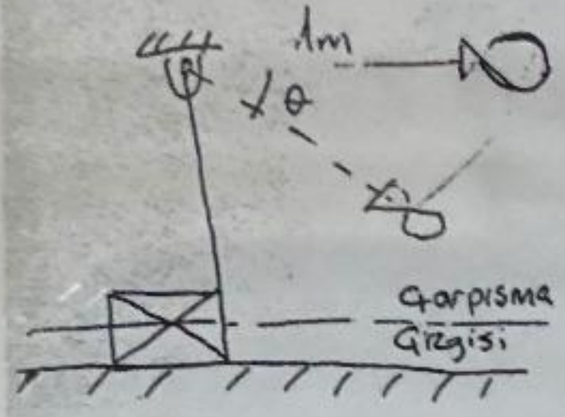
$0 = 1,03 \cdot \omega_{AB}^2 - 16,26$

$(\omega_{AB})_2 = 3,97$ rad/s

$v^2 = v_0^2 + 2.a$

$v^2 = 0 + 2.4$

c.3)



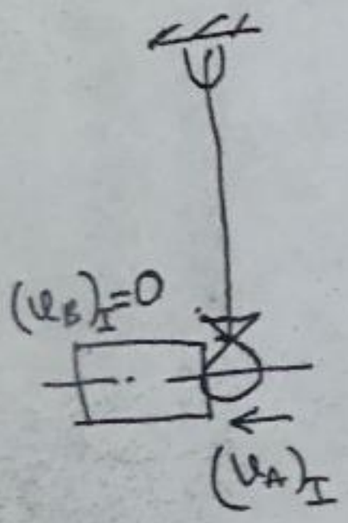
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Enerjinin Korunumu

$\theta = 0^\circ \quad T_0 + V_0 = T_I + V_I$

$0 + 0 = \frac{1}{2}\left(\frac{30}{9,81}\right) \cdot (v_A)_I^2 - 30.1$

$(V_A)_I = 4,43$ m/s



Momentumun Korunumu

A ve B'nin çarpışmadan hemen sonra sola doğru gittiğini düşünürsek;

Sisteme momentumun korunumu

$m_B \cdot (v_B)_I + m_A (v_A)_I = m_B \cdot (v_B)_2 + m_A \cdot (v_A)_2$

$0 + \frac{30}{9,81} \cdot 4,43 = \frac{90}{9,81} \cdot (v_B)_2 + \frac{30}{9,81} \cdot (v_A)_2$

$(v_A)_2 = 4,43 - 3 \cdot (v_B)_2$

Geri Dönme Katsayısı

$e = \frac{(V_B)_2 - (V_A)_2}{(V_A)_1 - (V_B)_1} \Rightarrow 0,5 = \frac{(V_B)_2 - (V_A)_2}{4,43 - 0}$

$(V_B)_2 = (V_A)_2 - 2,22$

$(V_A)_2 = 4,43 - 3 \cdot [(V_A)_2 - 2,22] \Rightarrow (V_A)_2 = -0,55$ m/s $= 0,55$ m/s $\rightarrow$

$(V_B)_2 = 1,66$ m/s

~~(1)/(2)~~ Enerji Kaybı; çarpışmadan hemen ~~sonra~~ önce ve hemen sonrası işe ve enerji ilkesini uygularsak

$\Sigma U_{1\to 2} = T_2 - T_1 = \left[\frac{1}{2}\left(\frac{90}{9,81}\right) \cdot 1,66^2 + \frac{1}{2} \cdot \left(\frac{30}{9,81}\right) \cdot 0,55^2\right] - \left[\frac{1}{2}\left(\frac{30}{9,81}\right) \cdot (4,43)^2\right]$

$\dots = -16,9$ J

c.4)

Diskin kütle mer

diskin saat yönün

Diskin sabit dön

$I_A = \frac{1}{2} m \cdot r^2 = \frac{1}{2} \cdot \left(\frac{200}{9,81}\right)$

İmpuls ve Momentum ilke

$\overset{+}{\rightarrow} \quad m \cdot (v_{Ax})_1 + \sum \int_{t_1}^{t_2} F_x \, dt =$

$0 + R_{Ax} \cdot 2 = 0$

$+\uparrow \quad m \cdot (v_{Ay})_1 + \sum \int_{t_1}^{t_2} F_y$

$0 + R_{Ay} \cdot 2 - 1$

$\overset{+}{\curvearrowright} \quad I_A \cdot \omega_1 + \sum \int_{t_1}^{t_2} M$

$0 + 6.2 + (5$

$T_2 = 115,104 N$

$v^2 = v_0^2 + 2.a.\Delta x$

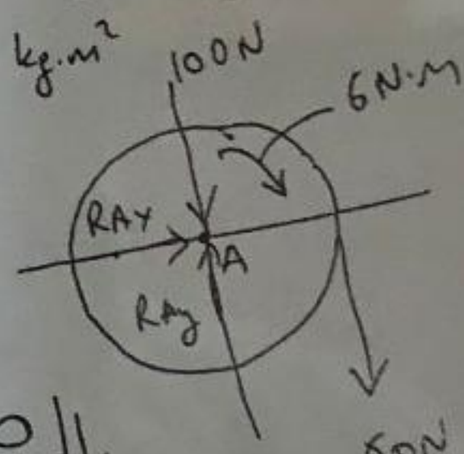
$v^2 = 0 + 2.4,054.2 \Rightarrow \underline{v = 4,03 \text{ m/s} \downarrow}$

c.4)

Diskin kütle merkezi hareket etmez; ancak yükleme diskin saat yönünde dönmesine neden olur.

Diskin sabit dönme eksenine göre atalet momenti

$\bar{I}_A = \frac{1}{2} m.r^2 = \frac{1}{2}.\left(\frac{200}{9,81}\right)(0,25)^2 = 0,319 \text{ kg.m}^2$



İmpuls ve Momentum ilkesi

$\overset{+}{\rightarrow} \quad m.(v_{Ax})_1 + \sum \int_{t_1}^{t_2} F_x.dt = m.(v_{Ax})_2$

$0 + R_{Ax}.2 = 0 \Rightarrow \underline{R_{Ax} = 0}$

$+\uparrow \quad m.(v_{Ay})_1 + \sum \int_{t_1}^{t_2} F_y.dt = m.(v_{Ay})_2$

$0 + R_{Ay}.2 - 100.2 - 50.2 = 0$

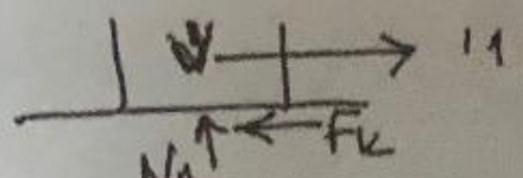
$\underline{R_{Ay} = 150 \text{ N}}$

$\overset{+}{\curvearrowright} \quad I_A.\omega_1 + \sum \int_{t_1}^{t_2} M_A.dt = I_A.\omega_2$

$0 + 6.2 + (50.0,25).2 = 0,319.\omega_2$

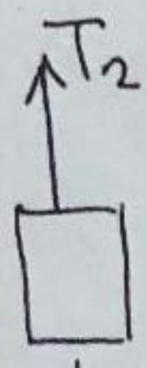
$\underline{\omega_2 = 116 \text{ rad/s}}$

z. Ayrıca, mafsaldaki tepki kuvvetinin bileşenleri nedir?
Momentum İlkesi) (25 puan)



$+\uparrow \sum F_y = m.a_y$; $N_A - 15.(9,81) = 0$ $N_A = 147,15\,N$

$\overset{+}{\rightarrow} \sum F_x = m.a_x$; $T_1 - 0,3.(147,15) = 15.a$

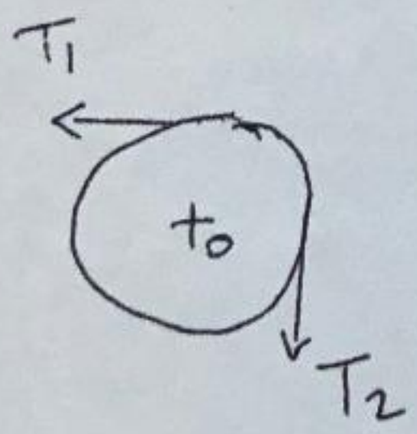


$+\uparrow \sum F_y = m.a_y$ $T_2 - 20.(9,81) = -20.a$

$a = r.\alpha \Rightarrow \alpha = \frac{a}{r} = \frac{a}{0,1} = 10.a$

$I_o = \frac{1}{2} m.r^2 = \frac{1}{2}.5.(0,1)^2 = 0,025\ kg.m^2$

$\sum M_o = I_o.\alpha$ $T_1.0,1 - T_2.(0,1) = -0,025.1$



$$\begin{aligned} T_1 - 0,3.(147,15) &= 15.a \\ T_2 - 20.(9,81) &= -20.a \\ T_1 - T_2 &= -2,5.a \end{aligned}$$

ortak çözülürse $a = 4,054\ m/s^2$

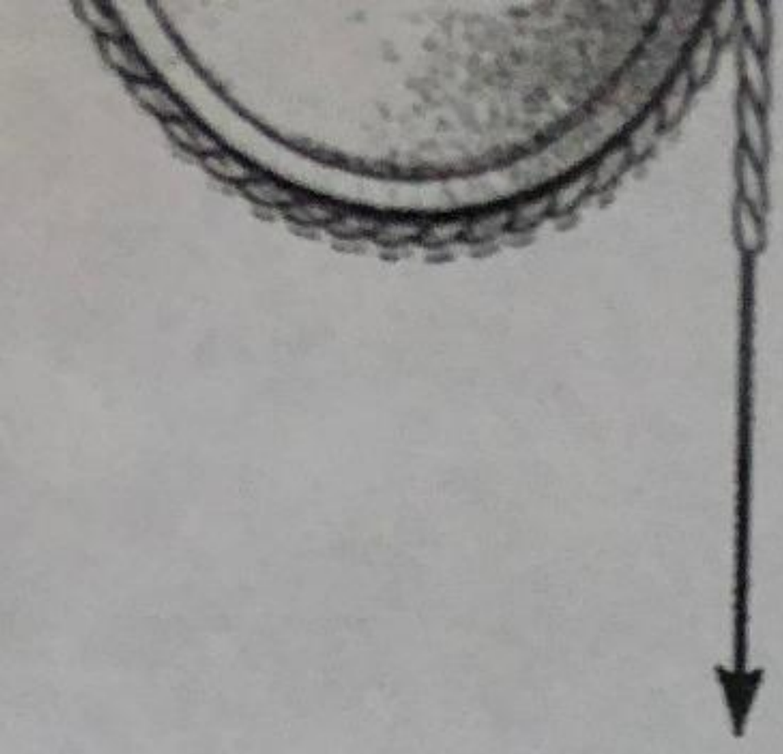
$T_1 = 104,967\,N$

$T_2 = 115,104\,N$

$v^2 = v_o^2 + 2.a.\Delta x$

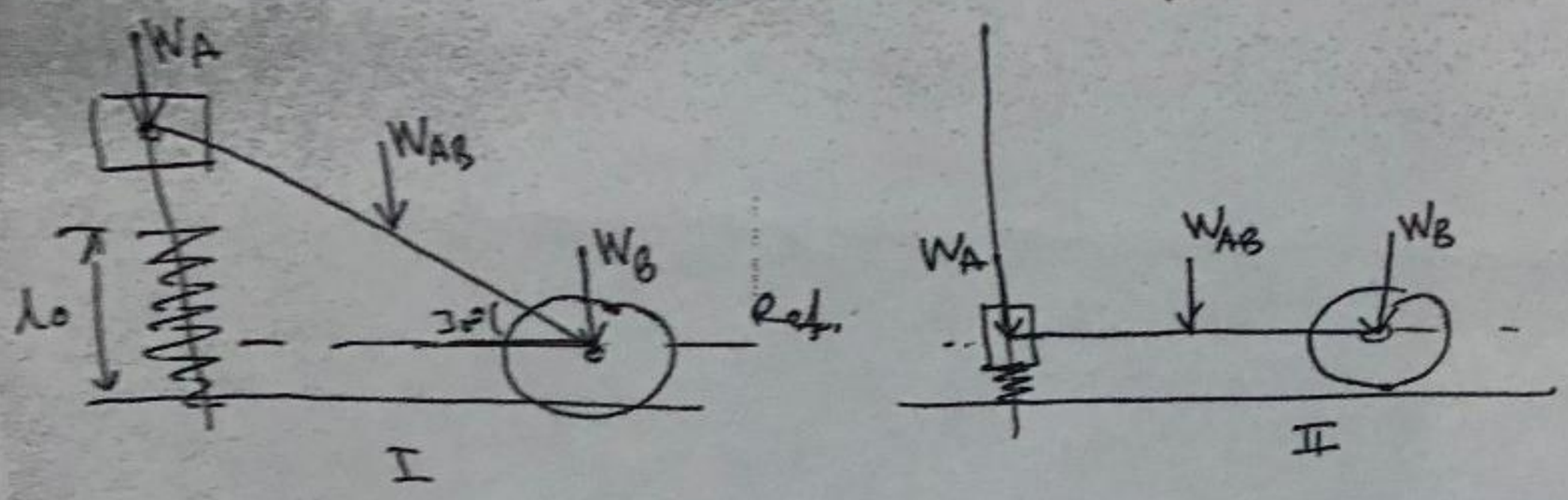
$v^2 = 0 + 2.4,054.2 \Rightarrow v = 4,03\ m/s \downarrow$

halden harekete
belirleyiniz. Ayrıca
(İmpuls ve Moment



$V_I = W_A \cdot y_{G_A} + W_{AB} \cdot y_{G_{AB}} + 0$

$V_I = 19,62 \cdot 0,375 + 49,05 \cdot 0,188 = 16,58$ J



$V_{II} = \frac{1}{2} k \cdot \Delta x^2 = 0,5 \cdot 100 \cdot (0,08)^2 = 0,32$ J.

$I_G = \frac{1}{12} \cdot m_{AB} \cdot l_{AB}^2 = 5 \cdot (0,75)^2/12 = 0,234$ kg·m²

$T_2 = \frac{1}{2} m_{AB} (v_G)_2^2 + \frac{1}{2} I_G (\omega_{AB})^2 + \frac{1}{2} m_A \cdot (v_A)_2^2$

$T_2 = \frac{1}{2} 5 \cdot (0,375 \cdot \omega_{AB})^2 + \frac{1}{2} \cdot 0,234 \cdot (\omega_{AB})^2 + \frac{1}{2} \cdot 2 \cdot (0,75 \cdot \omega_{AB})^2$

$T_2 = 1,03 \cdot \omega_{AB}^2$

Enerjinin Korunumu

$\Delta T = T_{II} - T_I = 1,03 \cdot \omega_{AB}^2$

$\Delta V = V_2 - V_1 = 0,32 - 16,58 = -16,26$ J

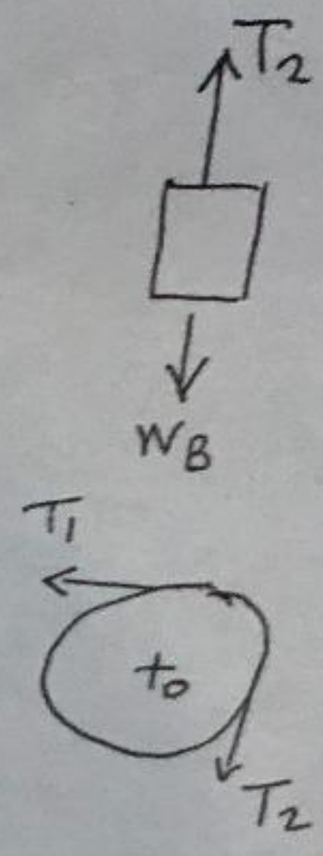
$0 = \Delta T + \Delta V$

$0 = 1,03 \cdot \omega_{AB}^2 - 16,26$

$(\omega_{AB})_2 = 3,97$ rad/s

$+\uparrow \Sigma$

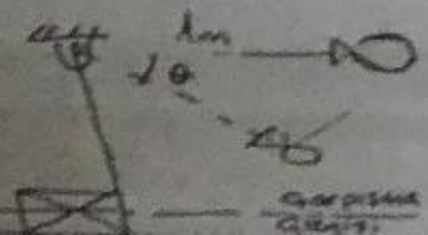
$\pm \rightarrow \Sigma$



$v^2 = v_0^2 + 2 \cdot$

$v^2 = 0 + 2 \cdot 4$

c.3)



Bu problem merkezcil cisimsel problemi ders Torkarın kütüğü cisimsel nedeniyle önceki hızını hemşlamak gerekir.

Enerjinin Korunumu

c.4)

Dişin katle merkezi
dişin saat yönünde